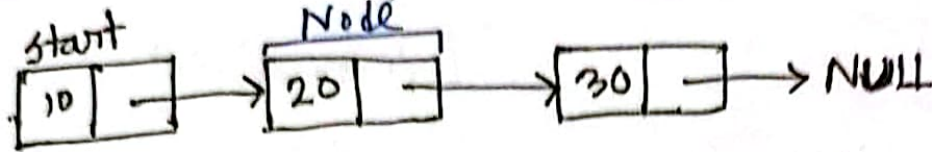
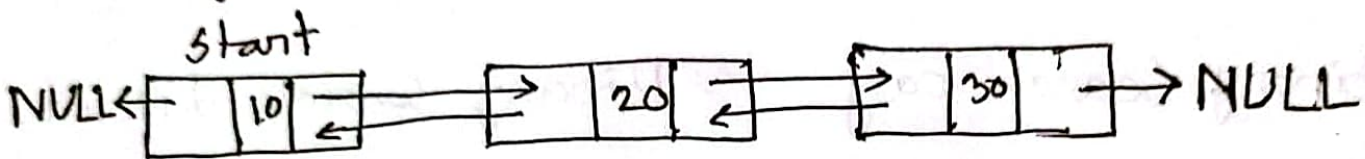


Different LinkedLists

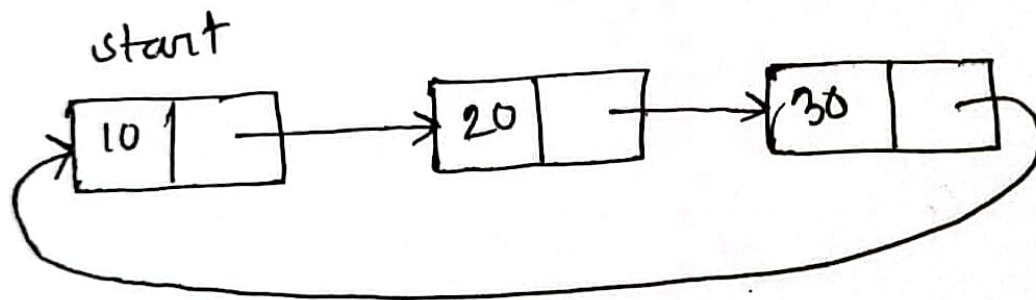
Linear or Singular LinkedList:



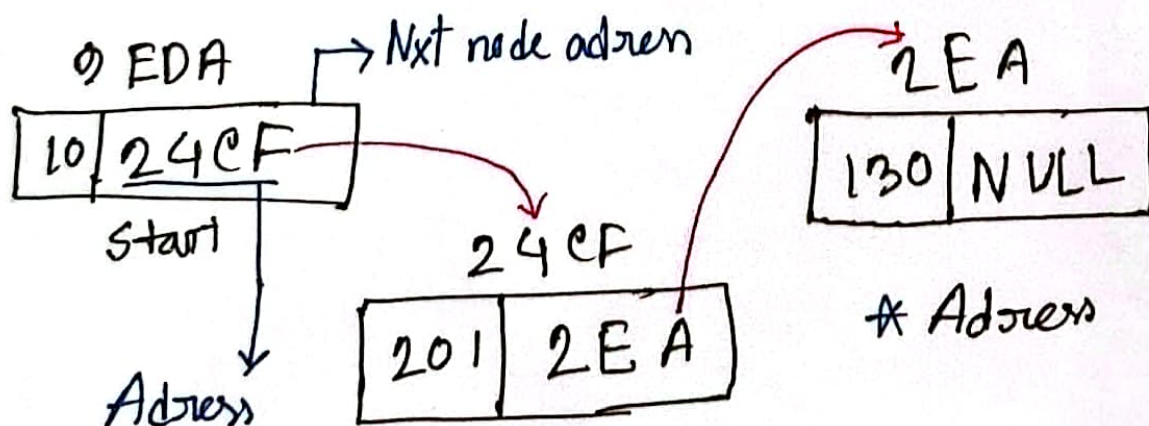
Doubly or Two-way LinkedList:



Linear circular linkedlist:

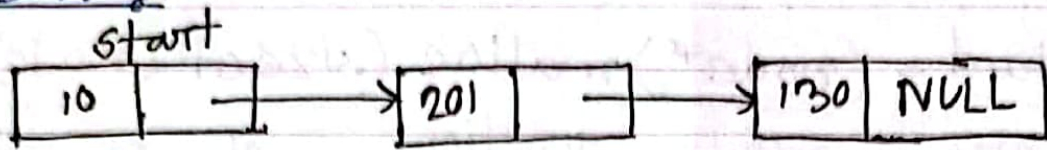


Memory allocation in LinkedList:

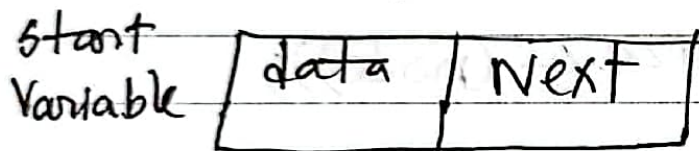
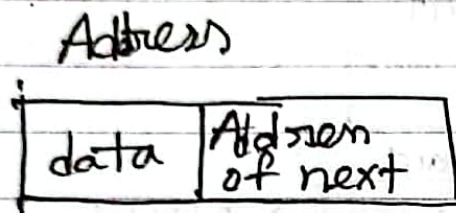


* Address Hexa-decimal.

Diagram :



Each node :



Variable Declaration :

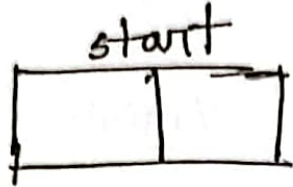
```

struct list {
    int Data ;
    struct list *next ;
};
// new datatype
// ↑
// new variable
typedef struct list node ;
int main() {
    node *start ;
}
  
```

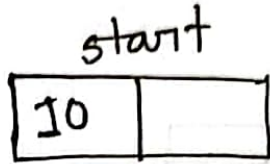

Statements.

1. start = (node*) malloc (size of (node));

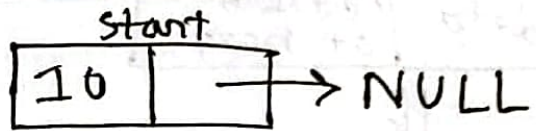
box name data typ function size of node



2. start → data = 10



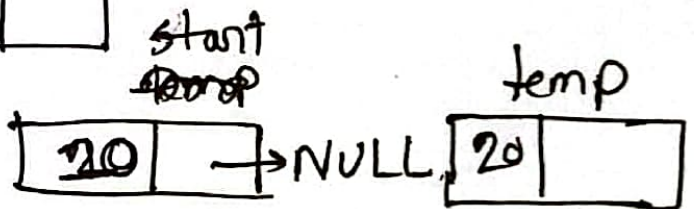
3. start → next = NULL



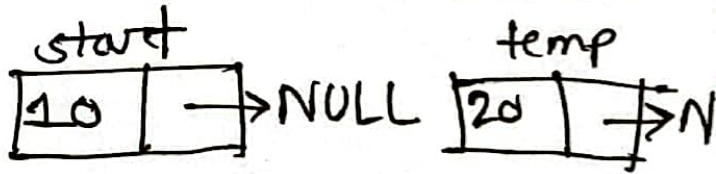
4. temp = (node*) malloc (size of (node));



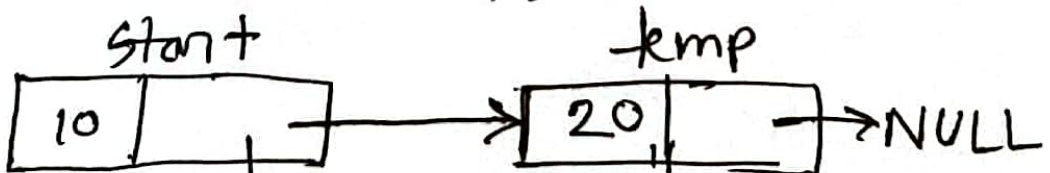
5. temp → data = 20



6. temp → next = NULL

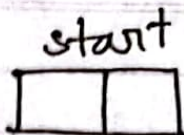


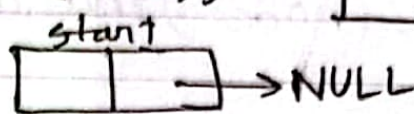
7. start → next = temp;

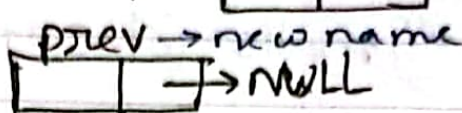


NULL Auto remove

* start box ke Next ke field temp box ke address par hai.

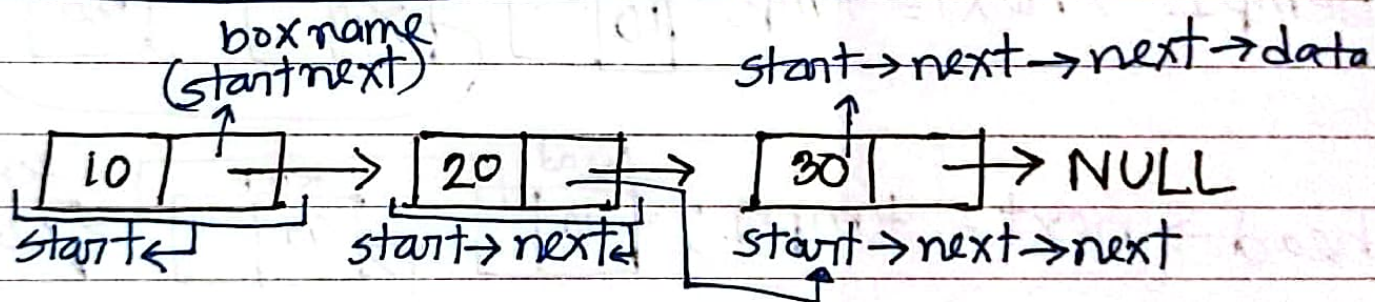
$start = (node^*) malloc (size\ of\ (node)) \rightarrow$ 

$start \rightarrow Next = NULL;$ 

$prev = start;$ 

```
for (i = 10; i <= 30; i = i + 10) {  
    temp = (node^*) malloc (size of (node));  
    temp  $\rightarrow$  data = i;  
    temp  $\rightarrow$  next = NULL;  
    prev  $\rightarrow$  next = temp;  
}
```

Statements for the linked list.



$start \rightarrow data = 10;$

$start \rightarrow next \rightarrow data = 20;$

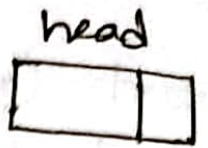
$start \rightarrow next \rightarrow next \rightarrow next = NULL$

$start \rightarrow next \rightarrow next \rightarrow data = 30;$

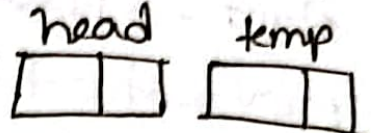
Exam: ***

1. Draw diagram for each of the statements given below:

$\text{head} = (\text{node}^*) \text{malloc}(\text{size of}(\text{node}));$



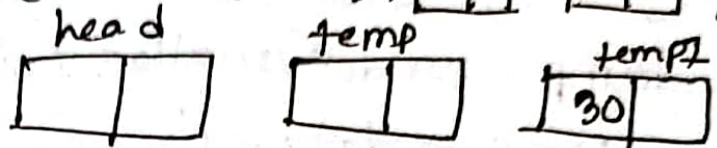
$\text{temp} = (\text{node}^*) \text{malloc}(\text{size of}(\text{node}));$



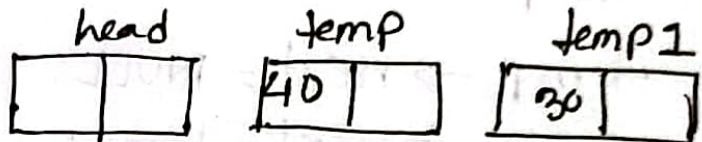
$\text{temp1} = (\text{node}^*) \text{malloc}(\text{size of}(\text{node}));$



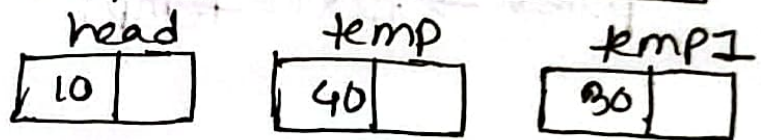
$\text{temp1} \rightarrow \text{data} = 30;$



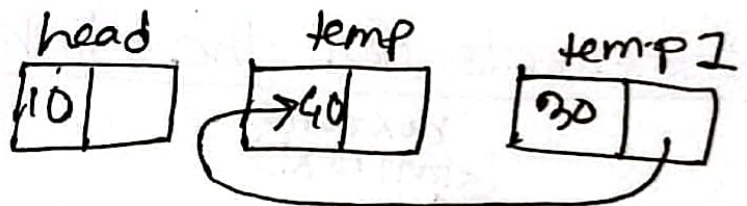
$\text{temp} \rightarrow \text{data} = 40;$



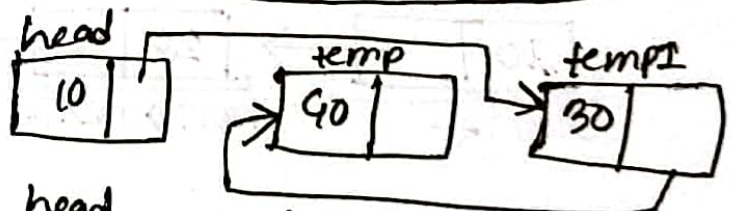
$\text{head} \rightarrow \text{data} = 10;$



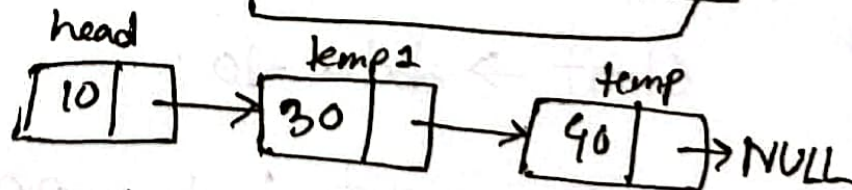
$\text{temp1} \rightarrow \text{next} = \text{temp};$



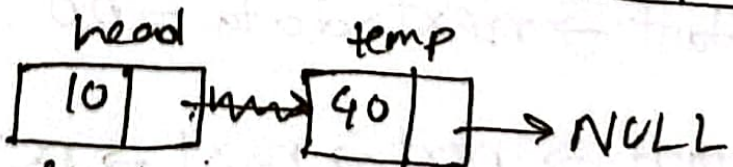
$\text{head} \rightarrow \text{next} = \text{temp1};$



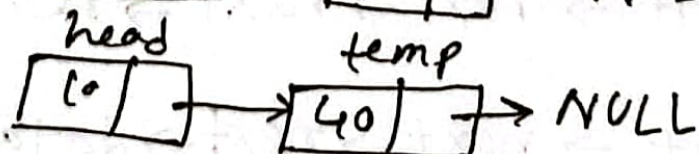
$\text{temp} \rightarrow \text{next} = \text{NULL};$



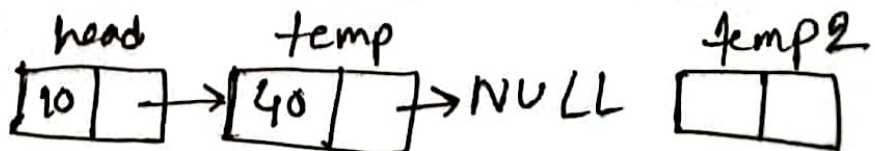
$\text{free}(\text{temp1});$



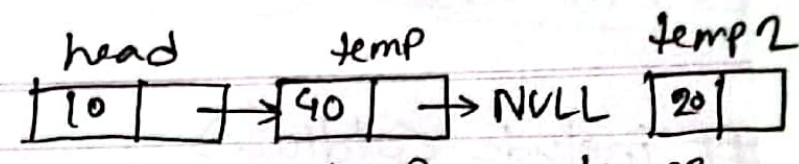
$\text{head} \rightarrow \text{next} = \text{temp};$



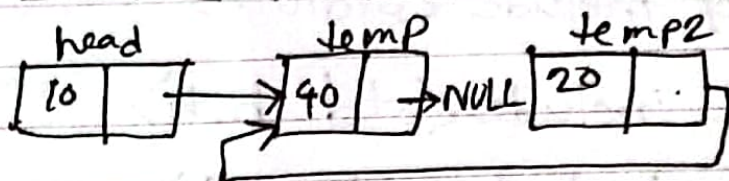
$\text{temp2} \rightarrow \text{data} = 20$



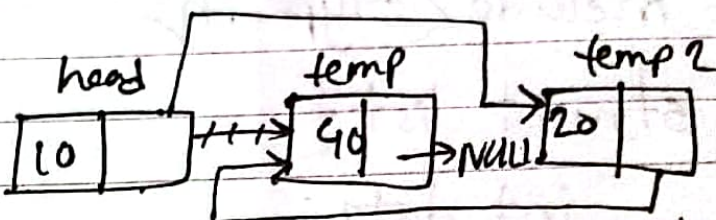
temp 2 → data = 20 ;



temp 2 → next = temp



~~temp~~ head → next = temp 2



Final Linked List :

